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Developing user-centered system design guidelines for explainable AI: a systematic literature review

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Abstract
The rapid advancement of AI technology has led to increasingly complex and opaque systems, creating a critical need for explainable AI (XAI) that enhances transparency and user trust. Despite extensive research on XAI methods and applications, the user-centered design (UCD) process for XAI remains fragmented and unclear. This systematic review analyzes 27 studies from 2020 to 2024 to develop a comprehensive framework for user-centered XAI system design (UCXAISD). We identify five key stages: contextual inquiry, explanation needs identification, XAI method selection, user interface design, and evaluation and refinement. Our framework aligns with traditional UCD processes while incorporating specialized elements for XAI, including user-centricity, transparency, and actionability. For each stage, we provide evidence-based guidelines derived from the literature. The framework serves as a blueprint for developing XAI systems that balance technical sophistication with user needs.

Keywords Explainability · Human-computer interaction · User-centered design

1 Introduction

In recent years, the field of Explainable Artificial Intelligence (XAI) has garnered significant research interest, driven by rapid advancements in AI technology. As various AI systems are being developed for different applications, a critical challenge has emerged: many of these systems, particularly those based on complex models, are inherently opaque or “black-box” in nature. This opacity poses a significant barrier to their utility and widespread adoption. XAI systems, capable of providing useful explanations about AI systems, hold the potential to greatly enhance the value and trustworthiness of AI technologies.

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Researchers have extensively studied the design of XAI systems. Most XAI research has focused on developing various explanation methods, ranging from traditional approaches like decision trees (Breiman et al. 1984) to modern techniques such as input gradients and sensitivity analysis (Selvaraju et al. 2016; Ribeiro et al. 2016; Lei et al. 2016). The field has also seen the emergence of diverse methodologies including generalized additive models (Caruana et al. 2015), procedural explanations (Singh et al. 2016), falling rule lists (Wang and Rudin 2015), exemplar-based approaches (Kim et al. 2014; Frey and Dueck 2007), and decision sets (Lakkaraju et al. 2016). Furthermore, another significant area of XAI design research focuses on domain-specific use cases, with applications spanning various sectors including clinical support (Antoniadi et al. 2021; Chen et al. 2022), financial markets (Yeo et al. 2023; Černevičienė and Kabašinskas 2024), legal systems (Ebers 2020; van Otterlo and Atzmueller 2021), and autonomous vehicles (Madhav et al., 2022; Malik et al., 2024; Atakishiyev et al. 2024).

While these technical advancements are crucial, parallel research has also been conducted on the user-centered design (UCD) of XAI systems. UCD aims to create systems that are intuitive, accessible, and aligned with user needs and expectations. It is characterized by elements such as user-centricity, transparency, actionability, consistency, real-time feedback, interactivity, iterative design and testing, cultural and ethical considerations, contextualization, and usability metrics (Green and Labrecque 2023; Ehsan et al. 2023; Swamy et al. 2023; Panigutti et al. 2023).

The UCD approach plays a pivotal role in the development of effective XAI systems as these systems fundamentally serve as the interface between AI and human users, facilitating human-AI collaboration. The UCD process for XAI systems must be tightly coupled with the AI system design process, ensuring that each step is informed by relevant design knowledge. Recent research in this area has focused on various aspects, including user-centered evaluation of XAI (He et al. 2023) and user needs elicitation (Cheligeer et al. 2022; Liao and Varshney 2021).

While several recent systematic reviews have examined aspects of UCD for XAI from different perspectives, they do not provide a holistic framework for the user-centered XAI system design process (UCXAISD). Raulino et al. (2024) provided a systematic mapping of UCD integration in machine learning and big data applications, highlighting usability factors such as perceived utility and learnability. Naveed et al. (2024) conducted a comprehensive review of empirical evaluations in XAI, emphasizing the lack of rigorous user-centered assessments and proposing a structured evaluation framework. Jung et al. (2025) focused on design recommendations for user-centered explanation interfaces in clinical decision support systems, underscoring the challenges in healthcare XAI implementations. Al-Ansari et al. (2024) performed a systematic literature review on user-centered evaluation of XAI, identifying gaps in evaluation methodologies and proposing a six-pillar framework for assessing XAI explainability.

However, despite these advancements, the UCD process for the design of XAI systems is not well understood. This gap in understanding can be attributed to several factors. One reason is that existing research studies on the UCD of XAI systems are fragmented and have not been synthesized into a cohesive whole. While literature reviews have been conducted, they did not focus on the UCD process for XAI system design. It is worth noting that the process of UCXAISD likely shares commonalities with generic system design processes, such as the user interface (UI) design process described by Yen and Davis (2019). However, the unique elements and characteristics of the UCXAISD process have not been elucidated.

This fragmentation of knowledge has led to several challenges. The lack of knowledge integration has resulted in the unavailability of well-organized design knowledge. For instance, currently, no design standards or guidelines similar to ISO 13,407 (1999) and ISO 9241 (2010) for UI design are available for XAI system design. This lack of understanding and knowledge organization hinders the design of sound XAI systems and consequently hampers the utility of AI systems.

To address these critical gaps, the objective of the current study is to review relevant literature and provide an integrated understanding and knowledge. Specifically, the current study seeks to address the following research questions:

- RQ 1: What are the distinctive user-centered elements and developmental stages that characterize UCXAISD processes?
- RQ 2: What evidence-based guidelines can be synthesized to effectively implement UCXAISD principles across each developmental stage of XAI systems?

By answering these questions, this study aims to contribute to the development of a more cohesive and structured approach to UCXAISD.

2 Background

UCD, rooted in human-computer interaction and software engineering, emphasizes considering user needs, preferences, and limitations throughout system development (Juárez-Ramírez, 2017). The UCD process comprises five interconnected stages (Yen and Davis 2019; Pushpakumar et al. 2023; Ramadhan et al. 2022): Research for understanding user needs and deployment context, Conceptual Design for defining system structure and functionality, Prototyping for creating tangible system representations, User Testing for identifying usability issues, and Iterative Refinement for continuously improving based on feedback. In the human-computer interaction (HCI) domain, these stages are implemented through four fundamental UCD elements: Visual Design (layout, color scheme, typography), Interaction Patterns (navigation, gestures, animation), Accessibility (WCAG compliance, assistive technologies, readability), and Feedback Mechanisms (error messages, confirmation dialogs, user surveys).

Building upon these UCD foundations, UCXAISD emerges as a domain-specific application that integrates UCD principles with XAI and human-AI interaction requirements for more effective and user-friendly AI systems. UCXAISD emphasizes key elements identified by Green and Labrecque (2023): user-centricity for prioritizing end-user needs by understanding AI system interaction context (Ehsan et al. 2023), transparency for providing clear decision-making explanations without technical jargon (Swamy et al. 2023), and actionability for enabling informed decisions through highlighted features and directly applicable insights (Panigutti et al. 2023). The framework also incorporates consistency for building trust and reliability (Swamy et al. 2023), real-time feedback and interactivity for dynamic environments (Swamy et al. 2023; Panigutti et al. 2023), cultural and ethical considerations for ensuring fairness, and contextualization for tailoring explanations to specific applications and user roles (Panigutti et al. 2023) (Fig. 1).

3 Methods

This systematic literature review followed the methodology outlined by Tranfield et al. (2003) and adhered to the PRISMA guidelines (Moher et al. 2015). The review focused on identifying key elements and stages in the XAI system design process, as well as practical design guidelines and knowledge.

We conducted searches in Scopus and ArXiv using the following search string (accessed in May 20th of 2024): ("explainability" OR "explainable AI" OR "explanation") AND ("user interface" OR "human-computer interaction" OR "user-centered design") AND ("design" OR "guidelines"). Scopus was chosen for its comprehensive coverage of peer-reviewed literature, while ArXiv was included to capture emerging research not yet indexed in traditional databases.

The initial search yielded 1,535 records (1,437 after removing duplicates). The screening process consisted of four stages (see Fig. 2):

- Initial screening (1,437 records): Basic eligibility criteria included English language only, publication years 2020–2024 (chosen due to XAI's significant research traction beginning around 2020), and publication types limited to journal articles, conference papers, reviews, and book chapters (Records excluded: 98).

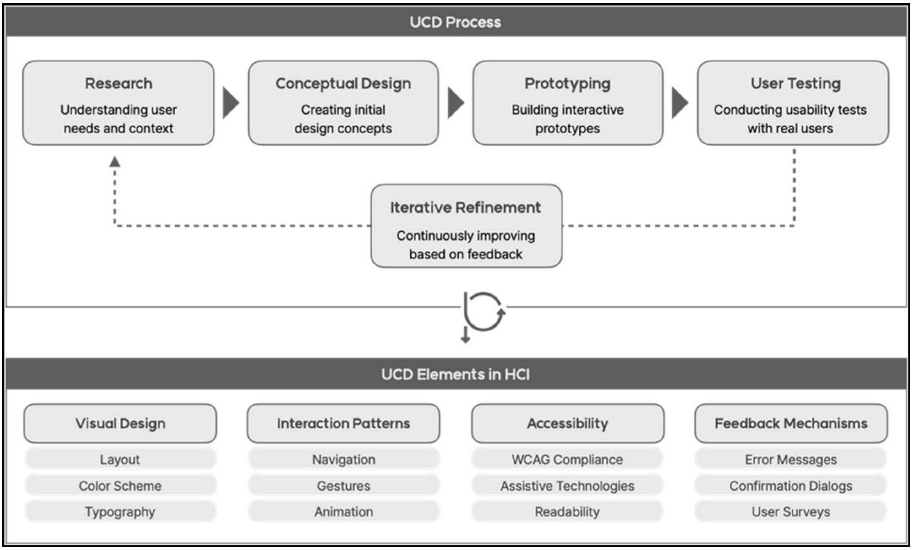

Fig. 1 UCD Process and Elements in HCI. The framework shows the traditional five-stage UCD process (Research, Conceptual Design, Prototyping, User Testing, and Iterative Refinement) supported by four fundamental HCI elements (Visual Design, Interaction Patterns, Accessibility, and Feedback Mechanisms). This traditional framework serves as the foundation for developing the specialized UCXAISD process, which incorporates additional XAI-specific considerations such as transparency, actionability, and user trust in AI decision-making contexts

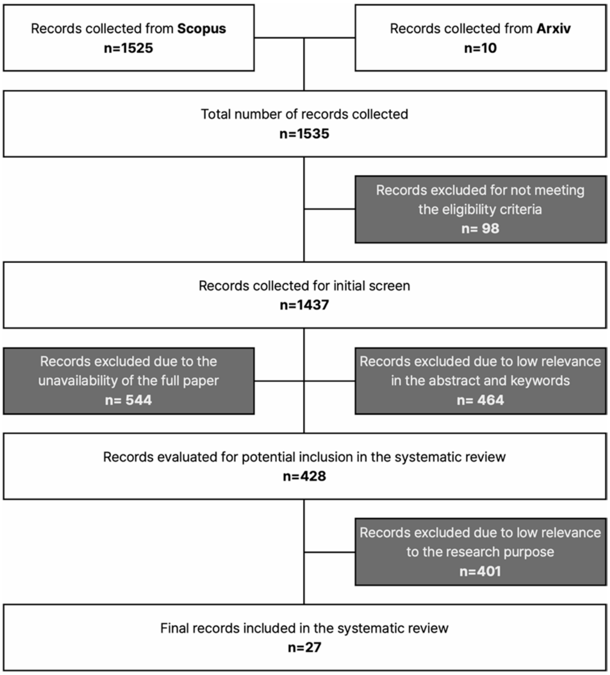

Fig. 2 PRISMA Flow Diagram for Systematic Literature Review Search Strategy

- Full-text assessment (1,437 records): Evaluated full text availability of all records (Records excluded: 544).
- Relevance screening (892 records): Reviewed abstracts and keywords for relevance to UCXAISD (Records excluded: 464).
- Detailed evaluation (428 records): Conducted in-depth assessment of alignment with research focus on user-centered explanation design for XAI (Records excluded: 401).
- Final included records: 27 (22 from Scopus, 5 from ArXiv).

Following the completion of the systematic selection process based on PRISMA guidelines, the final set of studies was subjected to a thematic synthesis to extract key insights relevant to UCXAISD. This process involved systematically coding findings across studies, identifying common themes, and synthesizing best practices for user-centered XAI system design.

For RQ1 (key elements and stages), we identified and categorized design process components and their relationships, and for RQ2 (practical guidelines), we extracted and synthesized design recommendations and implementation knowledge. The extracted studies were analyzed through an inductive thematic synthesis approach, where two independent coders systematically identified recurring themes related to user-centered design in XAI. A coding framework was developed iteratively, grouping insights into core design elements such as explanation types, user needs, and interface considerations. To ensure consistency, coding discrepancies were discussed and resolved with additional research assistants before finalizing the thematic structure. Practical guidelines and recommendations were extracted to inform implementation practices. Additionally, we recorded study limitations and suggestions for future research to provide context for the current state of knowledge in the field.

Two researchers independently coded the contents and themes within the papers. Their results were compared and discussed with additional research assistants to ensure consistency and reliability in the analysis. Figure 2 provides an overview of the search strategy and screening process.

The final dataset comprised 19 journal articles and 8 conference papers, with publication years distributed as follows: 2024 (6), 2023 (10), 2022 (8), 2021 (2), and 2020 (1).This distribution indicates growing research interest in UCXAISD.

Post-review addition: Following reviewer feedback, we incorporated Holzinger et al. (2011) and Del Ser et al. (2024) to strengthen our discussion on user-centered design principles in XAI systems. While Holzinger et al. (2011) predates our original inclusion criteria (2020–2024), it was deemed foundational for contextualizing user interaction frameworks in AI-driven systems, particularly its three-level perspective (micro, meso, and macro) for effective workflow integration. The inclusion of Del Ser et al. (2024) enhances our analysis of counterfactual explanation approaches and their impact on user trust. With these additions, our final dataset comprises 29 records: 21 journal articles and 8 conference papers, providing a more comprehensive foundation for our proposed UCXAISD framework.

4 Findings

Our systematic literature review addresses two primary research questions through the analysis of 27 selected studies. From this analysis, we propose the UCXAISD process illustrated in Fig. 3. This process comprises five sequential stages: Contextual Inquiry, Explanation Needs Identification, XAI Method Selection, UI design, and Evaluation and Refinement. The framework is designed to serve as a comprehensive blueprint for AI developers, designers, and researchers in developing XAI systems that adhere to UCD principles (Fig. 3).

To facilitate practical implementation, we provide detailed guidelines and actionable insights for each stage of the process. These guidelines are derived from our systematic analysis of the literature, which is summarized in Table 1's concept matrix. This matrix maps how each analyzed study contributes to specific stages and elements of our framework, providing a structured overview of the theoretical and practical foundations underpinning our proposed UCXAISD process (Table 1).

4.1 Contextual inquiry

The first stage of the UCXAISD process is contextual inquiry. It aims to understand the complete context in which the XAI system will operate by gathering comprehensive information about users, their work environment, and the AI system's role in their tasks. Based on the analysis of the selected research studies, we contend that contextual inquiry comprises three elements: User Profiling, Task Analysis, and AI Model Assessment. The three elements in combination ensure that the XAI system design is grounded in a thorough understanding of the users, their tasks, and the AI model's capabilities. In what follows, each element is described along with practical knowledge and insights relevant to it (Table 2).

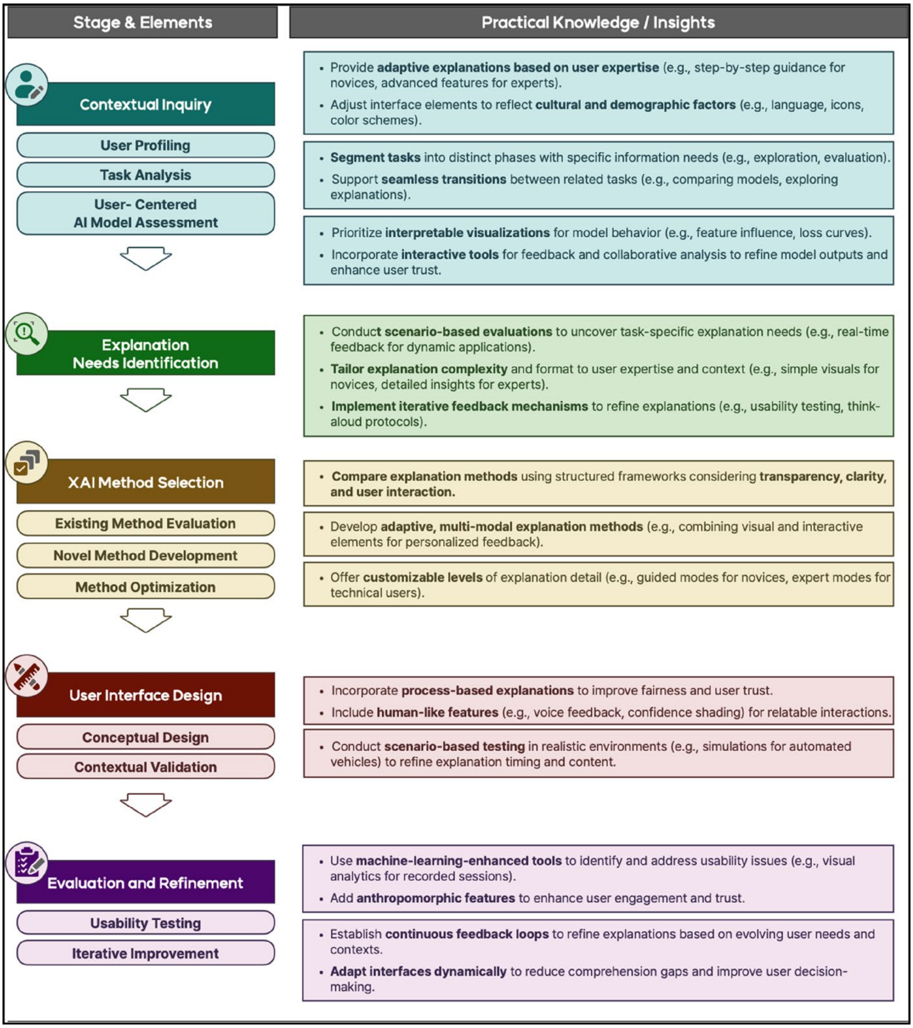

Fig. 3 Proposed UCXAISD Process Framework. This five-stage framework extends traditional UCD to address XAI-specific challenges: Contextual Inquiry (understanding users and AI capabilities), Explanation Needs Identification (determining explanation requirements), XAI Method Selection (choosing appropriate techniques), UI Design (creating effective interfaces), and Evaluation and Refinement (testing and improvement). Iterative feedback loops ensure explanations are both technically sound and user-centered

4.1.1 User profiling

User profiling is the process of collecting and analyzing information about users to understand their characteristics, preferences, behaviors, and needs (Thomas et al. 2024). Nine of the reviewed studies conducted distinct user profiling approaches through various contexts and methodologies, revealing important insights for real-world application.

Table 1 Concept matrix for literature analysis

Stage	Contextual Inquiry			Explanation Needs Identification	XAI Method Selection		UI Design		Evaluation and Refinement	
Elements	User Profiling	Task Analysis	User-Centnered AI Model Assessment		Existing Method Evaluation	Novel Method Development	Conceptual design	Contextual Validation	Usability Testing	Iterative Improvement
Alsswey et al. (2022)	X									
Alvarado et al. (2022)	X							X		
Alt et al. (2024)	X		X	X		X				
Cabour et al. (2022)				X	X					
Chou et al. (2022)					X					
Fan et al. (2020)									X	
Grafe et al. (2022)										X
Guesmi et al. (2022)					X					
Hoque et al. (2023)				X	X			X		
Huang et al. (2023)	X									
Kim et al. (2023)		X		X		X		X		
Li et al. (2024)				X						
Liu et al. (2022)			X		X					
Meng et al. (2022)		X		X		X				
Morrison et al. (2023)	X			X						
Müller-Birn et al. (2021)							X			
Muntwiler and Eppler (2023)										X
Nguyen et al. (2024)				X	X					
Sepich et al. (2023)	X			X	X				X	
Song et al. (2023)							X			
Staes et al. (2024)	X			X	X					
Thomas et al. (2024)	X				X		X		X	

Table 1 (continued)

Stage	Contextual Inquiry			Explanation Needs Identification	XAI Method Selection		UI Design		Evaluation and Refinement	
Elements	User Profiling	Task Analysis	User-Centnered AI Model Assessment		Existing Method Evaluation	Novel Method Development	Conceptual design	Contextual Validation	Usability Testing	Iterative Improvement
Weber and Hußmann (2022)	X							X		X
Weith and Matt (2023)							X			
Weitz et al. (2021)							X		X	
Wijekoon et al. (2023)		X		X		X				
Yada et al. (2023)	X			X						

Table 2 Synthesis of studies contributing to context inquiry in XAI system design

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
Contextual Inquiry	User Profiling	Thomas et al. (2024)	Active Learning Systems places significant emphasis on the necessity of comprehending the disparate user interaction requirements, particularly the impact of varying degrees of system transparency on the establishment of trust in automated decision-making processes.	Places significant emphasis on comprehending disparate user interaction requirements	• To accommodate diverse user expertise levels, XAI systems should incorporate a layered explanation approach, providing step-by-step guidance with concise, visual explanations and default configurations for novice users, while enabling advanced settings for expert users to explore in-depth model behavior, control feature prioritization, and interact directly with model parameters. User-centered design approaches that carefully consider device capabilities, interface usability, and user context can significantly improve adoption of new information systems across all stakeholder levels in clinical settings (Alt et al. 2024; Nguyen et al. 2024; Yada et al. 2023; Holzinger et al. 2011). • Cultural profiling should be leveraged to adapt interface elements such as language, iconography, and layout to align with cultural expectations, using color schemes and symbols that resonate culturally, and incorporating larger fonts, simplified navigation paths, and alternative input methods for older users to improve user satisfaction, accessibility, and interaction quality across varied demographic groups (Alsswey et al., 2022; Alvarado et al. 2022; Huang et al., 2023).

Table 2 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
					• In privacy-sensitive contexts, the level of detail in explanations should be tailored based on the user's role, with clinicians requiring in-depth explanations for diagnostic tasks, administrative staff needing higher-level insights for quick decision-making, and group-based systems implementing explanation formats that display consensus levels, source credibility, and reasoning traceability to support collaborative decision-making and enhance individual understanding and collective trust in the system's outputs (Staes et al., 2024; Morrison et al. 2023; Thomas et al. 2024).
		Nguyen et al. (2024)	Case studies examined how different user groups interpret hierarchical explanations in XAI systems, finding that users' technical expertise significantly influenced their ability to understand and interact with system behavior trees. The study documented varied interpretation patterns across different user expertise levels.	Demonstrated specific interpretation patterns across expertise levels, showing how technical background affects ability to understand complex XAI systems	
		Staes et al. (2024)	This study conducted iterative user profiling through case studies in privacy-sensitive domains, revealing that medical staff and clinicians require different levels of explanation detail based on their roles. The study emphasized how domain expertise shapes explanation needs in privacy-preserving ML contexts.	Provided role-based explanation requirements in privacy-sensitive domains through iterative user profiling in case studies	

Table 2 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
		Alvarado et al. (2022)	A comprehensive investigation was conducted to ascertain the influence of cultural dimensions on user UI preferences, with a particular focus on Arab users. The findings revealed the presence of discernible cultural patterns in how users interpret explanations and interact with interfaces. This underscores the significant impact of cultural backgrounds on user needs and satisfaction in XAI systems.	Conducted comprehensive investigation revealing discernible cultural patterns in explanation interpretation and interface interaction	
		Morrison et al. (2023)	The study analyzed collective user behavior in group recommender systems, identifying unique user profiling needs for collective decision-making scenarios. It documented how group dynamics affect individual user preferences and system interaction patterns.	Analyzed collective user behavior in group recommender systems, identifying unique profiling needs for collective decision-making	
		Yada et al. (2023)	This study conducted empirical user testing with participants using different XAI techniques, documenting user interaction patterns and visualization preferences based on domain expertise. The study compared LIME and Grad-CAM approaches, revealing that interpretation patterns vary significantly between technical and non-technical users with different visual preferences and comprehension rates.	Empirically compared LIME and Grad-CAM visualization preferences	

Table 2 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
		Alt et al. (2024)	Researchers compared static and interactive interfaces for explaining AI decisions in robot program optimization, profiling two distinct user groups: AI novices and experts. The study revealed experts required more advanced control features and detailed explanations, while novices needed simpler interfaces with guided workflows and default settings. Study highlighted importance of adapting interface complexity to user expertise level.	Presented quantitative demonstration of expertise-based interface requirements in robotics	
		Alsswey et al. (2022)	The study highlights the importance of incorporating cultural preferences in design components such as language, font, color, layout, and icons to align with users' cultural and traditional expectations. This approach ultimately increases user engagement and acceptance.	Established importance of culturally-aligned design elements	
		Huang et al. (2023)	The age-related decline in cognitive and physical abilities has a considerable impact on the usability of digital devices. Older adults often encounter difficulties when attempting to navigate complex layouts, read small text sizes, and interact with digital interfaces, which collectively impede their ability to engage effectively with e-health platforms.	Identified specific interaction limitations requiring alternative input methods	
		Holzinger et al. (2011)	The study demonstrated that mobile health applications significantly improved hospital workflows across all levels – patients found the system highly usable regardless of age, medical staff saved substantial time, and hospital management achieved meaningful cost savings while simultaneously enhancing data quality and patient empowerment.	Pioneered three-level perspective (micro/user, meso/task, macro/environment) for clinical UCD	

Table 2 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
	Task Analysis	Wijekoon et al. (2023)	The research demonstrated that interactive XAI dialogue systems benefit from behavior trees with varying granularity levels to handle different user needs and explanation strategies dynamically. The findings emphasized the importance of supporting multiple explanation intents (transparency, trust, debugging) while allowing users to pose follow-up questions or indicate disagreement, leading to more effective explanation experiences.	Pioneered dynamic support for multiple explanation intents	• Task analysis should segment XAI interactions into distinct phases (understanding, exploration, and evaluation), with each phase having specific information requirements and interaction patterns based on user goals. For instance, model comparison tasks require initial performance overview, followed by detailed feature exploration and explanation analysis (Meng et al. 2022; Kim et al. 2023).
		Kim et al. (2023)	The task analysis of autonomous vehicle interface interactions identified distinct user tasks related to building situational awareness, with users needing to understand both immediate vehicle perception and potential risk scenarios. The study found that users performed better at awareness-building tasks when provided with perception state information, while attention-based tasks required more cognitive effort and interpretation time, leading to slower task completion.	Identified cognitive effort differences in awareness-building tasks	• Analysis tasks should be structured to support both high-level goals (like understanding model behavior) and low-level operational needs (like examining specific feature impacts), while maintaining clear relationships between these task levels through coordinated interface components (Meng et al. 2022; Wijekoon et al., 2023).
		Meng et al. (2022)	Through detailed task analysis of model comparison workflows, the study identified three primary task categories: performance comparison tasks (comparing metrics and classification results), feature analysis tasks (exploring feature importance and distributions), and explanation comprehension tasks (understanding model decision-making). These tasks were found to be highly interconnected, with users frequently switching between tasks during analysis sessions and requiring synchronized views to maintain context across different analytical perspectives.	Demonstrated need for synchronized views to maintain context	• System designers should identify and support natural transitions between different analysis tasks, enabling users to seamlessly move between examining model performance, exploring explanations, and evaluating system trustworthiness while maintaining context across these transitions (Kim et al. 2023; Meng et al. 2022).

Table 2 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
					• Task workflows should adapt dynamically to user needs, supporting both structured paths for common analysis sequences and flexible exploration for unexpected insights or follow-up questions, while maintaining consistent support for core analysis tasks (Wijekoon et al., 2023; Kim et al. 2023).
	User-Centered AI Model Assessment	Alt et al. (2024)	The study emphasized structured AI model assessment through dataset evaluation, model training metrics, and interactive optimization visualizations to aid user understanding and decision-making. Layer-wise relevance propagation (LRP) was used to explain input feature impacts on model predictions.	First implementation of layer-wise relevance propagation in robotics UI	• AI model assessment should prioritize clear, interpretable visualizations that facilitate user understanding of model behavior and help identify data inconsistencies or anomalies. Presenting results in an accessible format ensures that outputs align with user expertise and domain-specific needs (Alt et al. 2024; Liu et al. 2022)
		Liu et al. (2022)	This study on visual analytics for detecting, investigating, and annotating anomalies in multivariate time series (MTV) highlights the importance of integrating human expertise into the AI model assessment process. It emphasizes how anomaly detection workflows benefit from human-machine collaboration to interpret, annotate, and refine ML outputs. Key findings include the necessity for models to present interpretable results and support collaborative user analysis to improve anomaly detection outcomes	Demonstrated value of collaborative annotation and refinement	• Assessment tools should allow domain experts to annotate and provide feedback on model outputs directly. This interactive capability helps refine model assessments over time, enhancing the system's ability to distinguish between legitimate and problematic outputs through iterative expert input (Alt et al. 2024; Liu et al. 2022) • A collaborative framework should be implemented in which multiple users can assess, discuss, and adjust the AI model's interpretations. This promotes comprehensive evaluations, harnessing collective expertise to refine model behavior and increase trustworthiness. (Liu et al. 2022)

The studies emphasized the importance of adaptive profiling methods to cater to the diverse needs of different user groups. Huang et al. (2023) identified age-specific constraints among older adults, highlighting the necessity for alternative input methods and shorter, focused interactions. Alt et al. (2024) noted the contrasting requirements of AI novices and experts, with the former benefiting from simplified interfaces and the latter from advanced control features. Staes et al. (2024) underscored the role of domain expertise in shaping profiling needs, revealing that medical staff and clinicians have varying requirements for explanation detail based on their roles. Particularly in clinical applications, as demonstrated by Holzinger et al. (2011) in their seminal work, user-centered design must always follow a three-level perspective: microlevel (user), mesolevel (task), and macrolevel (environment). This multi-dimensional approach is essential for effective workflow reengineering in healthcare settings and for properly evaluating the effectiveness of mobile applications in clinical contexts. Nguyen et al. (2024) discovered that users' technical expertise influenced their comprehension of hierarchical explanations, while Yada et al. (2023) documented user interaction patterns and visualization preferences based on domain expertise. Moreover, cultural factors play a crucial role in user profiling. Alvarado et al. (2022) and Alsswey et al. (2022) stressed the importance of incorporating cultural preferences in design components to align with users' expectations. Morrison et al. (2023) examined group behaviors in recommender systems, identifying distinct profiling needs for collective decision-making. Thomas et al. (2024) highlighted the impact of system transparency on user trust in automated decisions within active learning systems.

Based on these findings, to accommodate diverse user expertise levels, XAI systems should adopt a layered explanation approach, offering step-by-step guidance with concise, visual explanations and default configurations for novice users, while providing advanced settings for expert users to explore in-depth model behavior, control feature prioritization, and interact directly with model parameters (Alt et al. 2024; Nguyen et al. 2024; Yada et al. 2023). User-centered design approaches that carefully consider device capabilities, interface usability, and user context can significantly improve adoption of new information systems across all stakeholder levels in clinical settings (Alt et al. 2024; Nguyen et al. 2024; Yada et al. 2023; Holzinger et al. 2011). Cultural profiling should be employed to adapt interface elements such as language, iconography, and layout to align with cultural expectations, incorporating color schemes and symbols that resonate culturally, and providing larger fonts, simplified navigation paths, and alternative input methods for older users to enhance user satisfaction, accessibility, and interaction quality across varied demographic groups (Alsswey et al., 2022; Alvarado et al. 2022; Huang et al., 2023). In privacy-sensitive contexts, the level of detail in explanations should be tailored based on the user's role, with clinicians requiring in-depth explanations for diagnostic tasks, administrative staff needing higher-level insights for quick decision-making, and group-based systems implementing explanation formats that display consensus levels, source credibility, and reasoning traceability to support collaborative decision-making and enhance individual understanding and collective trust in the system's outputs (Staes et al., 2024; Morrison et al. 2023; Thomas et al. 2024).

Despite these advances, the convergence of findings across multiple studies reveals a methodological pattern in current XAI research. While these studies consistently demonstrate the need for adaptive profiling, they primarily rely on static user categorization based on initial assessments rather than considering how user needs and expertise may evolve during system interaction. This observation suggests that current profiling approaches may not fully capture the dynamic nature of user-system relationships, indicating potential opportunities for more flexible user modeling approaches that can accommodate changing user needs and growing familiarity with AI systems over time.

4.1.2 Task analysis

Task analysis refers to a systematic examination of how users interact with XAI systems to accomplish their goals, helping designers create more effective user-centric interfaces. Three of the 27 studies reviewed in this study conducted task analysis in different contexts. Kim et al. (2023) analyzed on-road autonomous vehicle interactions to understand how users build situational awareness and process different types of explanations, finding that perception-based tasks were more efficiently completed than attention-based tasks. Meng et al. (2022) examined model comparison workflows to identify three interconnected task categories (performance comparison, feature analysis, and explanation comprehension), revealing how users switch between these tasks while maintaining analytical context. Wijekoon et al. (2023) investigated interactive XAI dialogue systems using behavior trees to understand how users engage with different explanation strategies and handle follow-up questions or disagreements. Alt et al. (2024) compared static versus interactive explanatory tools in robotics to understand how different interface modes support various user expertise levels and tasks.

Several practical guidelines for conducting task analysis during XAI system design were identified. XAI interactions should be segmented into distinct phases with specific information requirements and interaction patterns based on user goals (Meng et al. 2022; Kim et al. 2023). Analysis tasks should support both high-level goals and low-level operational needs, with clear connections through coordinated interface components (Meng et al. 2022; Wijekoon et al., 2023). System designers should identify and support natural transitions between different analysis tasks, allowing users to seamlessly move between examining model performance, exploring explanations, and evaluating system trustworthiness while maintaining context (Kim et al. 2023; Meng et al. 2022). Task workflows should adapt dynamically to user needs, supporting both structured paths for common analysis sequences and flexible exploration for unexpected insights or follow-up questions, while maintaining consistent support for core analysis tasks (Wijekoon et al., 2023; Kim et al. 2023).

Notably, a few studies reveal a tension between user needs for both structure and flexibility. While users benefit from structured explanation workflows, they also require the ability to deviate from prescribed paths when their specific needs demand it. This balance between guidance and autonomy challenges conventional UCD assumptions about linear task flows and suggests that XAI interfaces may need fundamentally different interaction paradigms.

4.1.3 User-centered AI model assessment

User-centered AI model assessment focuses on evaluating how well AI systems produce outputs that meet user needs and facilitate interpretation. Two of the 27 reviewed studies conducted AI model assessment in distinct contexts, providing insights and guidelines for XAI system design.

Alt et al. (2024) developed an Explanation UI (XUI) for robot program optimization, which allowed users to assess AI model training by visualizing metrics such as loss curves and employing explainability tools like layer-wise relevance propagation (LRP). This enabled users to understand which input features impacted the model's decisions, improving comprehension and decision-making during optimization. Liu et al. (2022) introduced the Multivariate Time Series Visualization (MTV) system to support AI model assessment through interactive visual tools for anomaly detection. The system facilitated user investigation, annotation, and interpretation of machine learning outputs, emphasizing the role of user input in refining models and enhancing accuracy through contextual feedback and collaborative analysis.

Based on these studies, several practical guidelines for AI model assessment during XAI system design were identified. Systems should include interpretable visualizations that make model behavior clear, such as loss curves and feature influence displays (Alt et al. 2024; Liu et al. 2022). User annotation capabilities should be incorporated to enable feedback, supporting continuous model refinement and reducing false positives (Liu et al. 2022; Alt et al. 2024). Collaborative analysis tools should be integrated to allow multiple users to share insights and contribute to comprehensive evaluations (Liu et al. 2022). Results should be presented in a contextually accessible manner that accommodates users of different expertise levels, ensuring clarity without sacrificing necessary detail for experts (Alt et al. 2024; Liu et al. 2022).

4.2 Explanation needs identification

Explanation needs identification aims to understand the types of explanations users require to effectively interact with XAI systems, enhancing user trust, comprehension, and informed decision-making. Twelve of the reviewed studies conducted explanation needs identification in various contexts (Table 3).

The studies employed a range of methods to identify explanation needs, including context-specific interviews, observations, and case studies. In clinical settings, Staes et al. (2024) found that medical staff require different explanation depths depending on their roles, highlighting the importance of tailoring explanations to specific user groups within a domain. Sepich et al. (2023) and Li et al. (2024) demonstrated that real-time, interactive explanations improve human-agent collaboration in dynamic environments like gaming and knowledge graph exploration, respectively. Furthermore, user expertise significantly influences explanation requirements. Alt et al. (2024) and Nguyen et al. (2024) showed that novice users benefit from simple, step-by-step guidance, while expert users in technical domains require detailed explanations and advanced control features for in-depth exploration. Especially in Alt et al. (2024)'s study, the transparency of AI-based decision-making revealed key differences between novice and expert users. Both groups demonstrated a high level of understanding of data provenance ($\mu_{novice}=4.125$, $\mu_{expert}=4.0$). However, novices struggled more with complex tasks such as output prediction (6.4), and both groups found it challenging to interpret the Layer-wise Relevance Propagation (LRP) plots (6.9), highlighting a need for improved explanation methods for intricate AI functionalities. Huang et al. (2023), Alvarado et al. (2022), and Yada et al. (2023) emphasized the importance of considering users' cultural and demographic factors, such as age and technical background, when determining explanation complexity and format. Additionally, scenario-based tests

Table 3 Synthesis of studies contributing to explanation needs identification in XAI system design

Stage	Study	Relevant findings	Major contribution	Practical knowledge/insights
Explanation Needs Identification	Sepich et al. (2023)	The objective of this study was to analyze human-agent teaming in video game interfaces with a view to identifying the explanation needs that arise in this context. The findings suggest that interactive, context-aware explanations are likely to facilitate user understanding and improve human-agent collaboration in complex tasks.	Pioneered analysis of explanation needs specific to human-agent teaming in dynamic gaming environments	• Context-specific interviews, observations, and case studies are employed to identify explanation needs that are unique to different roles and settings. In clinical contexts, medical staff require different explanation depths depending on their roles. Additionally, real-time, interactive explanations have been shown to improve human-agent collaboration in dynamic environments like gaming (Staes et al., 2024; Sepich et al. 2023; Li et al. 2024). • User expertise significantly influences explanation requirements. For novices, simple, step-by-step guidance is beneficial, while expert users in technical domains need detailed explanations and advanced control features for in-depth exploration. Users' cultural and demographic factors, such as age and technical background, should also inform explanation complexity and format (Alt et al. 2024; Nguyen et al. 2024; Huang et al., 2023; Alvarado et al. 2022; Yada et al. 2023).

Table 3 (continued)

Stage	Study	Relevant findings	Major contribution	Practical knowledge/insights
				• Scenario-based tests uncover specific explanation needs tied to tasks within certain applications. Autonomous vehicle users, for instance, benefit from situationally appropriate explanations for building trust, while knowledge graph users require guidance that simplifies complex queries for better understanding (Kim et al. 2023; Li et al. 2024; Thomas et al. 2024). • It is recommended that iterative feedback processes be implemented through the use of think-aloud protocols, usability testing, and game-based evaluations. This approach allows for the continuous capture and refinement of user explanations needs. This feedback allows for the modification of explanations to align with user preferences, facilitate comprehension, and address challenges in high-stakes or collaborative contexts, such as group recommendation systems (Morrison et al. 2023; Cabour et al. 2022; Weitz et al., 2021; Hoque et al., 2023).
	Nguyen et al. (2024)	A comprehensive survey of explainable interfaces was conducted, which revealed that user requirements for explanations vary based on user expertise and context of use. This highlighted the necessity for adaptive explanation methods tailored to different user profiles.	Provided first comprehensive survey of explainable interfaces across different expertise levels	

Table 3 (continued)

Stage	Study	Relevant findings	Major contribution	Practical knowledge/insights
	Staes et al. (2024)	Semi-structured interviews were conducted with oncologists to identify the explanation elements necessary for discussing machine learning-based prognostic information. This iterative design approach involved continuous feedback to refine interfaces, with a particular focus on the clarity of model outputs and their interpretability in the context of clinical practice.	Established iterative design methodology for clinical XAI interface development	
	Morrison et al. (2023)	A game-based method, designated "Eye into AI," was employed to assess user engagement with XAI techniques, including LIME, Grad-CAM, and feature visualizations. This approach underscored the significance of a user-centric evaluation for elucidating the efficacy of diverse explanation formats in facilitating comprehension.	Pioneered gamification for XAI technique evaluation	
	Yada et al. (2023)	The investigation of the detection of dark patterns in user interfaces using explanation techniques such as LIME and SHAP demonstrated the pivotal role of these explanations in assisting users in identifying manipulative patterns. This underscores the necessity for clarity and transparency in XAI interfaces.	Demonstrated explanation techniques' effectiveness in dark pattern detection	
	Alt et al. (2024)	The study found that incorporating explanation features tailored to different user expertise levels was crucial for user comprehension and effective interaction. This highlighted the necessity of adaptable explanations that meet the diverse needs of naive and expert users in technical domains.	Quantified comprehension differences between naive and expert users in technical domains	
	Wijekoon et al. (2023)	The authors proposed an interactive dialogue model for XAI systems, which employs behavior trees to formalize user explanation needs and tailor interactions. The study demonstrated the value of dynamic and personalized explanations in meeting various user requirements in real time.	Introduced dynamic and personalized real-time explanation framework	
	Kim et al. (2023)	The study examined the efficacy of visual explanations in highly automated vehicle interfaces, demonstrating that the provision of such explanations (e.g., pertaining to perception and attention states) during driving enhanced user trust and situational awareness. The findings indicated that the timing and modality of these explanations had a significant impact on the user experience, underscoring the importance of contextually appropriate explanations.	Presented quantitative study showing impact of explanation timing and modality on user experience	

Table 3 (continued)

Stage	Study	Relevant findings	Major contribution	Practical knowledge/insights
	Meng et al. (2022)	The study presented a visual analytics tool for comparing classification models, highlighting the importance of thorough feature analysis and clear explanations of model behavior. The results showed that effective model comparisons require explanations that clarify the reasoning behind model decisions and pinpoint differences between models.	Developed visual analytics approach for comparative understanding of classification models	
	Hoque et al. (2023)	The introduction of a visual analytics tool for data-centric AI has enabled users to interact with visual concepts and explanations on a large scale. This study demonstrated that user-friendly and interpretable explanation interfaces are essential for aiding non-expert users in understanding complex visual data models. The tool's interactive nature allows users to build, verify, and refine visual explanations, thereby demonstrating the necessity of user involvement for more effective model interpretation.	Demonstrated how visual analytics tools allow users to build, verify, and refine visual explanations	
	Li et al. (2024)	Knowledge graph (KG) users, both analysts and consumers, face challenges in querying due to the complex languages required, often necessitating specialized training. To address this, KG consumers expressed a need for user-friendly graphical interfaces and interim query results that simplify exploration and make insights more accessible to non-technical users.	Identified specific interface needs for non-technical knowledge graph users	
	Cabour et al. (2022)	The authors proposed an "explanation space" framework that links user requirements directly with XAI development. The framework places particular emphasis on aligning explanations with users' mental models and cognitive processes. This framework emphasizes the necessity of designing explanations that are tailored to specific user goals and work practices. It suggests that such alignment can facilitate more effective comprehension and practical use of XAI in decision-making contexts.	Presented comprehensive framework linking user requirements directly with XAI development	

have proven effective in uncovering specific explanation needs tied to tasks within certain applications. Kim et al. (2023) found that autonomous vehicle users benefit from situationally appropriate explanations for building trust, while Li et al. (2024) showed that knowledge graph users require guidance that simplifies complex queries for better understanding. Thomas et al. (2024) further emphasized the importance of tailoring explanations to specific application contexts. To ensure that explanations align with user requirements, the studies recommend implementing iterative feedback processes through the use of think-aloud protocols, usability testing, and game-based evaluations. Morrison et al. (2023), Cabour et al. (2022), Weitz et al. (2021), and Hoque et al. (2023) demonstrated that this approach allows for the continuous capture and refinement of user explanation needs, facilitating the modification of explanations to align with user preferences, improve comprehension, and address challenges in high-stakes or collaborative contexts, such as group recommendation systems.

The findings from these studies provide several practical guidelines for identifying explanation needs in XAI system design. First, designers should conduct context-specific interviews, observations, and case studies to identify unique explanation requirements for different roles and settings (Staes et al., 2024; Sepich et al. 2023; Li et al. 2024). Second, user expertise and cultural background should be considered when determining explanation complexity and format (Alt et al. 2024; Nguyen et al. 2024; Huang et al., 2023; Alvarado et al. 2022; Yada et al. 2023). Third, scenario-based tests should be employed to uncover specific explanation needs tied to tasks within certain applications (Kim et al. 2023; Li et al. 2024; Thomas et al. 2024). Finally, iterative feedback processes should be implemented to ensure that explanations align with user preferences and facilitate comprehension (Morrison et al. 2023; Cabour et al. 2022; Weitz et al., 2021; Hoque et al., 2023).

Despite the extensive research on explanation needs identification, a critical theoretical gap emerges from our analysis. Most studies focus on explicit user requirements gathered through interviews and observations, yet fail to address the well-documented phenomenon of users being unable to articulate their true needs until they experience different explanation approaches. This requirement articulation gap suggests that current identification methods may be systematically missing implicit user needs, potentially leading to suboptimal explanation designs.

4.3 XAI method selection

The third stage of the UCXAISD process, XAI Method Selection, aims to identify and implement the most appropriate explanation techniques that effectively address user needs while considering factors such as user expertise, task requirements, and context of use.

Before delving into the process of selecting appropriate XAI methods, it is essential to establish a clear typology of explanations. Explanation types in XAI can be broadly categorized based on their cognitive function, structure, and intended audience. Drawing on insights from the social sciences, Miller (2019) emphasizes that human-centered explanations tend to be contrastive (focusing on “why A instead of B”), selective (providing only the most relevant causes), and social (tailored to the explainee’s expectations). These insights suggest that effective XAI explanations must align with human reasoning patterns rather than purely technical descriptions.

Cabitza et al. (2023) further expand on this by introducing a structured typology of explanations in XAI, distinguishing between epistemic and pragmatic explanation types.

Epistemic explanations focus on formal, objective justifications of AI decisions, commonly aligned with algorithmic transparency. These include model-based explanations, which reveal the internal structure of an AI system (e.g., decision trees, feature importance rankings), and rule-based explanations, which provide human-readable logical conditions governing AI outputs (e.g., IF-THEN rules). In contrast, pragmatic explanations are centered on user cognition and decision-making, making them essential for real-world interpretability. These include counterfactual explanations, which illustrate how an output would change if input conditions were different (e.g., “Had X been 5 instead of 3, the model’s decision would have been Y instead of Z”), and example-based explanations, which compare the current decision to similar past cases or prototypical instances to improve user comprehension.

The distinction between these explanation types is crucial for XAI method selection, as different contexts and user groups require different forms of explanation. For example, counterfactual explanations (Wachter et al. 2017) are particularly valuable in domains where understanding alternative outcomes is critical, such as finance and healthcare. Example-based explanations (Kim et al. 2016) are useful when users benefit from concrete, relatable instances, such as in recommendation systems. Additionally, some XAI techniques combine multiple explanation types to enhance interpretability, such as hybrid approaches integrating visual case-based reasoning with statistical transparency metrics (Cabitza et al. 2023). Figure 4 illustrates how epistemic and pragmatic explanations differ in terms of structure and user accessibility in a hypothetical loan approval scenario.

Recognizing the diversity in explanation needs, the selection of XAI methods must be guided by a systematic evaluation of their effectiveness and suitability for different user groups. Based on the analysis of the selected nine research studies, XAI Method Selection

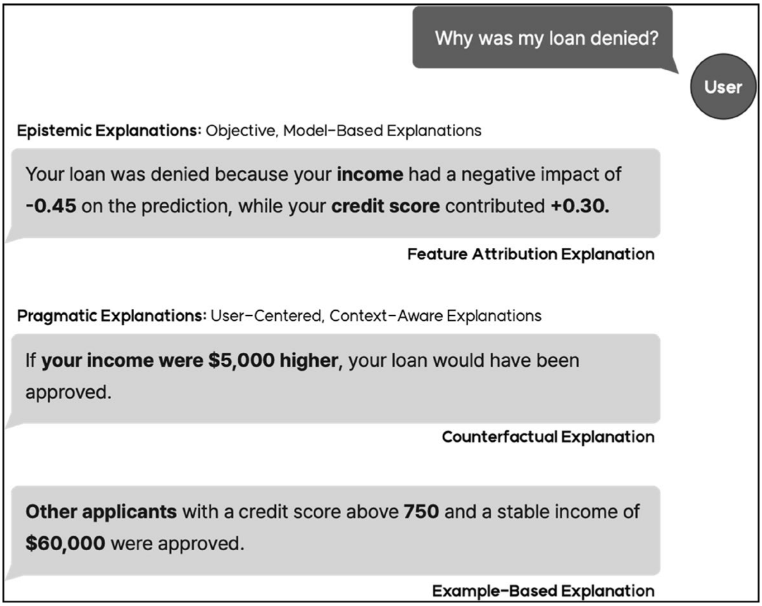

Fig. 4 Visualization of Epistemic and Pragmatic Explanations in a Hypothetical Loan Approval Scenario. Epistemic explanations provide formal, technical justifications (feature importance, model processes), while pragmatic explanations focus on user comprehension through counterfactuals and examples

comprises three elements: Existing Method Evaluation, Novel Method Development, and Method Optimization (Table 4).

4.3.1 Existing method evaluation

Existing Method Evaluation systematically assesses available explanation methods to determine their effectiveness and suitability for different users and contexts. Nine of the 27 reviewed studies conducted existing method evaluation in various settings. Chou et al. (2022) found that counterfactual algorithms excelled in providing actionable feature change recommendations, but noted challenges when multiple feature modifications were required. Counterfactuals are particularly valuable for enhancing user trust, as they enable users to familiarize themselves with unfamiliar processes by understanding the hypothetical input conditions under which the output changes. As Del Ser et al. (2024) demonstrate, this "what if?" approach not only promotes explainability but also significantly increases user confidence in the results by providing multi-faceted counterfactual explanations that balance plausibility, change intensity, and adversarial power. Sepich et al. (2023) showed that combining case-based explanations with visual examples enhanced human-agent collaboration. Thomas et al. (2024) discovered that incorporating uncertainty information in simplified visualizations improved user trust and decision-making. Nguyen et al. (2024) compared LIME and SHAP, finding that LIME's simpler approach was effective for basic understanding, while SHAP's depth better served technical users. Staes et al. (2024) revealed healthcare professionals' strong preference for interpretability-focused explanations. Hoque et al. (2023) achieved significant improvements in user comprehension by integrating visual explanations with algorithmic transparency (Fig. 5). Cabour et al. (2022) developed a structured "explanation space" framework for systematic comparison of diverse explanation approaches. Liu et al. (2022) established the advantages of interactive visual methods over traditional statistical approaches for time series analysis. Guesmi et al. (2022) explored recommender systems and highlighted how interactive model comparison could boost user comprehension and decision confidence.

The reviewed studies provided valuable insights for selecting appropriate explanation methods in XAI system design. First, systematic comparisons should be conducted across multiple dimensions, including algorithmic transparency, visual clarity, and user interaction capabilities, using structured frameworks to assess different approaches against specific user needs (Cabour et al. 2022; Nguyen et al. 2024; Liu et al. 2022). Second, evaluation should employ multiple criteria spanning technical accuracy, user comprehension, and practical applicability, with interactive evaluations comparing visualization techniques alongside algorithmic methods to identify complementary strengths (Hoque et al., 2023; Thomas et al. 2024; Guesmi et al. 2022). Third, methods should be evaluated within their intended context, with particular attention to domain-specific requirements and constraints (Staes et al., 2024; Sepich et al. 2023; Thomas et al. 2024). Finally, evaluations should consider both user expertise levels and task requirements, assessing immediate comprehension and long-term utility, particularly for complex explanations like counterfactuals. When generating counterfactual explanations, it is valuable to consider multiple criteria simultaneously, such as plausibility, change intensity, and adversarial power, to provide richer insights into model behavior and potential biases (Chou et al., 2022; Nguyen et al. 2024; Liu et al. 2022; Del Ser et al. 2024).

Different explanation methods align with specific user expertise levels and task contexts, making method selection crucial for effective XAI systems. Example-based explanations are highly suitable for novice users, as they rely on relatable comparisons to simplify AI decisions. Counterfactual explanations help intermediate users by allowing them to explore alternative outcomes, making them useful for decision-making in dynamic contexts. For expert users, feature attribution explanations and rule-based explanations provide detailed insights into the model's internal logic, offering transparency for AI developers, analysts, or auditors. These differences highlight the need to tailor XAI method selection to both user capabilities and task requirements to optimize system explainability and usability. Figure 6 illustrates how different explanation methods can be customized for users with varying expertise levels, using a hypothetical loan approval scenario. Each explanation type is justified in terms of its suitability for the respective user level, highlighting the trade-offs between interpretability, technical depth, and user comprehension.

4.3.2 Novel method development

Novel Method Development focuses on creating innovative explanation approaches when existing methods cannot fully address specific user or domain requirements. Two of the 27 reviewed studies contributed to novel method development in different contexts. Wijekoon et al. (2023) developed a behavior tree-based dialogue model that enables explanations to adapt dynamically based on user feedback and domain context (Fig. 7). This approach allows for real-time adaptation to changing user needs and varying situational demands, especially in high-stakes environments where user trust and safety are paramount. Kim et al. (2023) demonstrated that continuously providing perception information through semantic segmentation maps achieved the highest usability (78.7 ± 9.93) and trust (3.60 ± 0.80) compared to other explanation types. Additionally, when perception and attention information were selectively provided only in risky scenarios, participants reported improved situational awareness (5.27 ± 0.76) and reduced cognitive load (5.50 ± 2.32) compared to continuous provision of both. These findings highlight how dynamic, context-sensitive visual explanations can effectively enhance user trust, safety perception, and situational awareness in autonomous vehicle environments.

The development of these novel methods revealed important practical guidelines for XAI system design. Designers should focus on combining multiple modalities, such as visual feedback and interactive dialogue, to create comprehensive explanation experiences. This multi-modal approach has proven particularly effective in complex domains where users need immediate, contextually relevant feedback (Wijekoon et al., 2023; Kim et al. 2023). Additionally, novel methods should be designed with flexibility in mind, allowing for real-time adaptation to changing user needs and varying situational demands.

4.4 UI design

The UI Design stage, which is the fourth phase of the UCXAISD process, focuses on developing an interface that effectively conveys the outputs of AI systems. The goal is to ensure that users can comprehend, interact with, and have confidence in the system. Our analysis of the selected research studies suggests that UI design consists of two components: Conceptual Design and Contextual Validation. Each component is explained below, along

Table 4 Synthesis of studies contributing to XAI method selection in XAI system design

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
XAI Method Selection	Existing Method Evaluation	Chou et al. (2022)	This study conducted a systematic evaluation of counterfactual algorithms, highlighting their effectiveness in providing actionable insights and challenges in user interpretation, especially for complex cases requiring multiple feature changes.	Presented comprehensive evaluation of counterfactual algorithms for feature-change recommendations	• When evaluating XAI methods, it is recommended to conduct systematic comparisons across different dimensions - algorithmic transparency, visual clarity, and user interaction capabilities. Structured evaluation frameworks help assess the effectiveness of different approaches (e.g., LIME vs. SHAP, visual vs. statistical) in meeting specific user needs and task requirements (Cabour et al. 2022; Nguyen et al. 2024; Liu et al. 2022). • It is recommended that explanation methods are assessed using multiple criteria, including technical accuracy, user comprehension, and practical applicability. By conducting interactive evaluations that compare different visualization techniques in conjunction with algorithmic methods, complementary strengths and potential integration opportunities can be revealed (Hoque et al., 2023; Thomas et al. 2024; Guesmi et al. 2022). • It is essential to evaluate the efficacy of explanation methods within the context for which they are intended. This evaluation should take into account the specific requirements and constraints of the domain in question. In domains where the stakes are high, the evaluation criteria may differ from those used in exploratory applications. In such cases, particular attention should be paid to the interpretability and decision support capabilities of the method in question (Staes et al., 2024; Sepich et al. 2023; Thomas et al. 2024).

Table 4 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
					• Method evaluations should be structured around user expertise levels and task requirements. In order to assess method effectiveness, it is necessary to consider both immediate comprehension and long-term utility, particularly in the case of complex explanations such as counterfactuals. When generating counterfactual explanations, it is valuable to consider multiple criteria simultaneously, such as plausibility, change intensity, and adversarial power, to provide richer insights into model behavior and potential biases (Chou et al., 2022; Nguyen et al. 2024; Liu et al. 2022; Del Ser et al. 2024).
		Sepich et al. (2023)	The research found that interactive, context-aware methods, such as case-based explanations combined with visual examples, were particularly effective in enhancing user understanding and facilitating seamless human-agent collaboration.	Established effectiveness of case-based explanations with visual examples	
		Thomas et al. (2024)	It found that visual representations that combined simplicity with comprehensive uncertainty information—like shaded areas indicating prediction confidence—helped users better calibrate their trust and make informed decisions. Explanations that were overly technical or cluttered reduced user trust and engagement.	Demonstrated specific visual elements that enhance or reduce trust	
		Nguyen et al. (2024)	A comparative analysis of LIME and SHAP across different user groups revealed that LIME's simpler feature-based explanations were more effective for basic understanding, whereas SHAP's comprehensive analysis better served users with technical expertise.	Directly compared LIME and SHAP across different user expertise levels	
		Staes et al. (2024)	The study revealed that clinicians preferred explanations emphasizing interpretability over detailed technical explanations, highlighting the need for tailored communication in sensitive domains	Identified healthcare-specific explanation preferences	

Table 4 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
		Del Ser et al. (2024)	This study proposes a framework that uses GANs and multi-objective optimization to generate trustworthy counterfactual explanations for AI models by balancing plausibility, change intensity, and adversarial power, thereby providing more comprehensive insights into model behavior and revealing biases.	Pioneered balanced approach to counterfactual generation considering plausibility, change intensity, and adversarial power	
		Hoque et al. (2023)	The study evaluated the integration of visual explanations with algorithmic transparency in concept programming and found that this combination significantly enhanced user comprehension when compared to the individual methods.	Demonstrated quantifiable benefits of combining visual and algorithmic approaches	
		Cabour et al. (2022)	The study developed the "explanation space" framework that enables systematic comparison of diverse explanation methods, providing structured evaluation criteria for assessing the effectiveness of visual, textual, and hybrid approaches in different contexts.	Created structured evaluation criteria for visual, textual, and hybrid approaches	
		Liu et al. (2022)	A comparative analysis of visual analytics techniques with traditional machine learning explanations for time series data was conducted, which revealed that interactive visual methods facilitated more intuitive insights and superior anomaly detection compared to statistical approaches.	Presented comparative analysis of visual analytics vs. traditional ML explanations for time series data	
		Guesmi et al. (2022)	The exploration of user model visualization in recommender systems demonstrated that the interactive comparison of different models significantly enhanced user comprehension and decision confidence.	Established methodology for visualizing recommender system models	

Table 4 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
	Novel Method Development	Wijekoon et al. (2023)	This research proposed a unique interactive dialogue model using behavior trees to create adaptive, multi-shot explanations. This method supports personalized, real-time interaction, refining user engagement and trust by tailoring explanations dynamically based on user needs and domain specifics	Pioneered formalized approach to dynamic explanation adaptation using behavior trees	• Combining visual and interactive elements, such as real-time perception cues and behavior tree-driven dialogues, enhances user trust and comprehension by providing tailored, situation-specific explanations. This approach is particularly beneficial in complex, high-stakes environments like automated driving or technical operations, where immediate and personalized feedback is crucial for user confidence and safety (Wijekoon et al., 2023; Kim et al. 2023)
		Kim et al. (2023)	The study explored novel approaches in highly automated vehicles by developing visual explanation methods, including perception and attention-based visual cues. This innovation focused on timing and content to optimize user trust, safety, and situational awareness under real driving conditions	Presented quantitative study measuring effects of perception and attention-based visual cues on trust and safety	
	Method Optimization	Alt et al. (2024)	This study focused on optimizing explanation methods in industrial robotics by designing an adaptable user interface that provided two modes: a guided mode for novice users and an expert mode for advanced users. The guided mode simplified explanations, offering step-by-step assistance, while the expert mode allowed for more detailed, technical insights.	Pioneered dual-mode interface design optimized for robotics program optimization	• It is beneficial to permit users to customize their level of engagement in accordance with their expertise. By furnishing interactive visualization tools that facilitate comprehensive comparison of model behaviors, users are able to adapt explanations to align with their particular requirements and enhance comprehension across diverse user profiles. This methodology facilitates informed decision-making and assists in identifying areas for model refinement (Alt et al. 2024; Meng et al. 2022).
		Meng et al. (2022)	The study introduced an interactive visual analytics tool that enabled users to compare multiple classification models side-by-side. This tool optimized explanations by allowing users to switch between various visual representation types (e.g., feature importance plots and decision boundaries) to better understand and diagnose model behaviors	Developed novel approach to model comparison through dynamic visual representation switching	

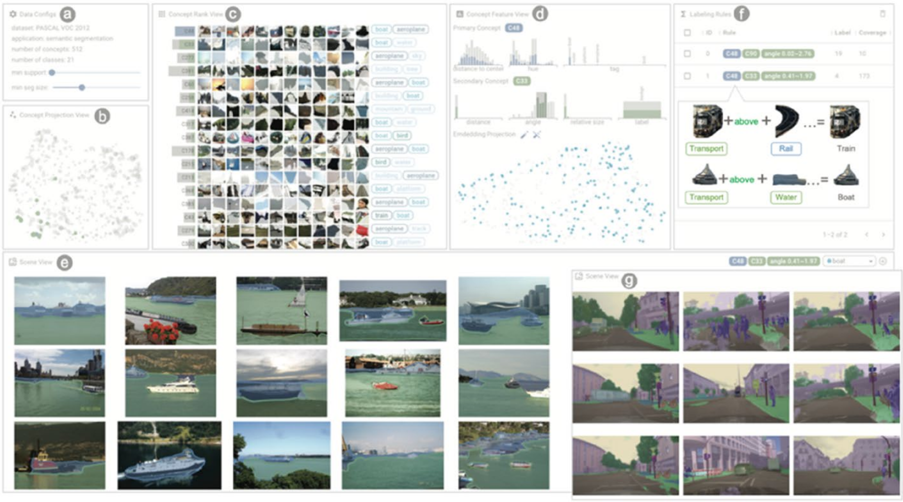

Fig. 5 Visual Concept Programming UI.
(Adapted from Hoque et al., 2023). This interactive interface combines visual explanations with algorithmic transparency, allowing users to create and refine visual explanations through direct manipulation

with relevant practical knowledge and insights. Table 5 offers a structured summary of the current research contributing to this stage, synthesizing the studies that are pertinent to UI design for XAI systems.

4.4.1 Conceptual design

Conceptual Design for XAI systems emphasizes creating explanations that precisely address user-specific needs and situational contexts. Weith and Matt (2023) revealed that process-based explanations, which offer insights into the steps or reasoning of the AI, are more effective than purely visual cues in enhancing fairness perceptions, particularly when personalized to reflect individual user profiles (Fig. 8). In Weith and Matt's (2023) study, process explanations significantly improved systemic fairness perceptions ($F(1, 227)=4.934$, $p<0.001$, $\eta^2 = 0.021$) compared to explanations that relied solely on visualizations, which showed no significant effect. Specifically, user-based process explanations had the greatest impact, enhancing distributive fairness ($F(3, 312)=13.583$, $p<0.001$) and informational fairness ($F(3, 312)=6.595$, $p<0.001$), resulting in a notable increase in fairness perceptions. Thomas et al. (2024) demonstrated that incorporating visual indicators, such as shaded areas to represent confidence levels, allows users to better interpret data uncertainty, fostering a stronger sense of trust in the AI's outputs. Weitz et al. (2021) showed that virtual agents with relatable, human-like features, such as spoken feedback and visual presence, can make interactions more intuitive and accessible, thereby increasing user trust and engagement. Müller-Birn et al. (2021) highlighted the value of iterative co-design, emphasizing that continuous refinement through user feedback allows explanations to be finely tuned to specific tasks and contexts. Song et al. (2023) proposed the use of Human Digital Twins (HDT) to dynamically adjust explanations in real time, leveraging personal and contextual data to deepen user engagement and relevance (Fig. 9).

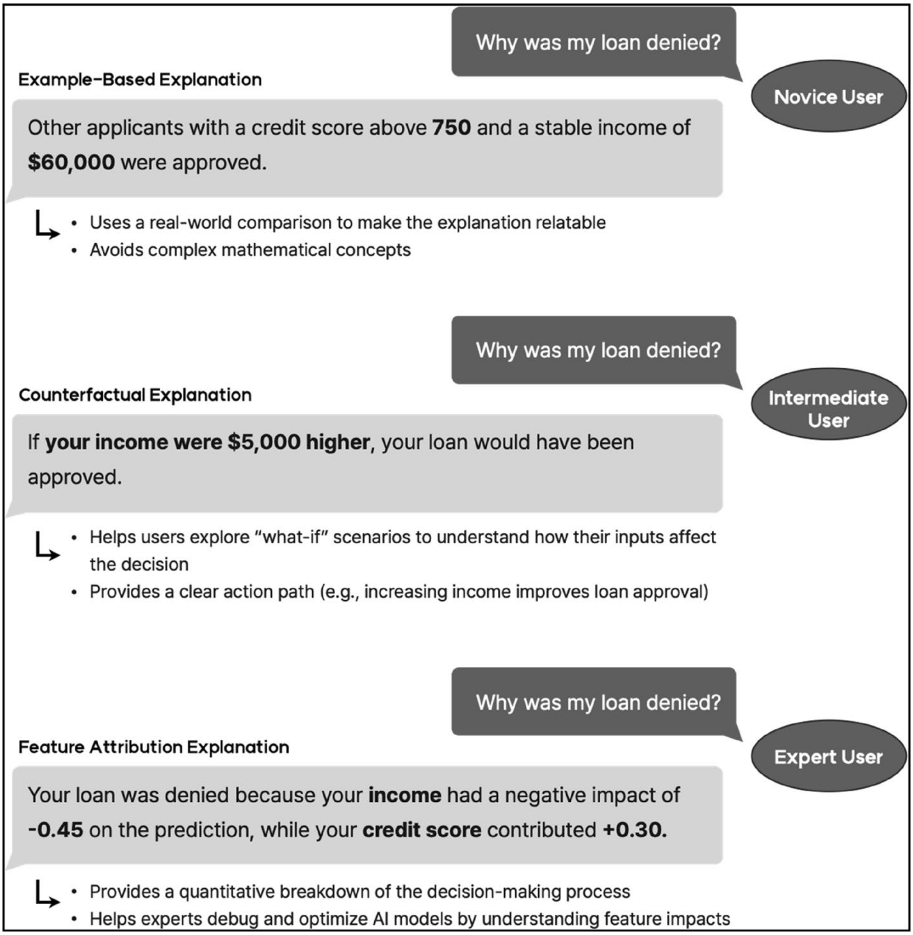

Fig. 6 Expertise-Based Explanation Variations in XAI: A Hypothetical Loan Approval Scenario. The illustration shows how explanation complexity should be tailored to user expertise levels: novice users receive simple visual explanations with plain language, intermediate users get moderate technical depth with confidence indicators, and expert users access comprehensive technical details including feature attribution scores

Several practical insights emerge from these studies for Conceptual Design. First, adopting an iterative co-design process is essential for refining interfaces based on real-world user feedback, ensuring that explanations align with specific contexts and dynamic user needs (Müller-Birn et al. 2021). Second, incorporating human-like features, such as voice feedback and adaptive visual cues like confidence shading, enhances trust by making complex information accessible and relatable (Weitz et al., 2021; Thomas et al. 2024). Finally, prioritizing process explanations, particularly in voice agent design, can improve fairness perceptions. Tailoring these explanations to individual user profiles and integrating real-time adjustments through HDTs ensures a personalized and equitable experience, fostering user confidence and satisfaction (Weith and Matt 2023; Song et al., 2023).

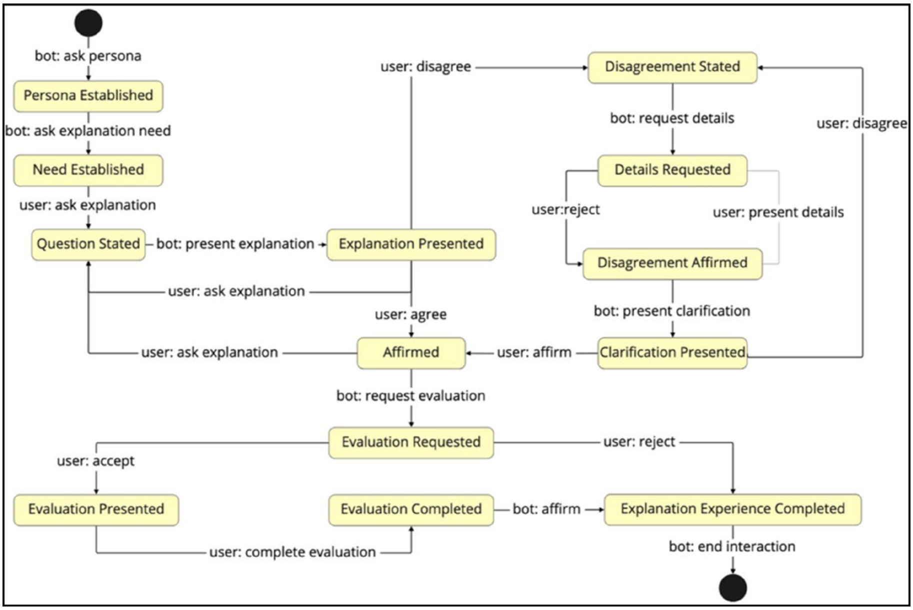

Fig. 7 Behavior Tree-Based Dynamic Dialogue Model for Adaptive XAI Explanations. (Adapted from Wijekoon et al., 2023). The hierarchical structure shows how behavior trees enable personalized, real-time explanations that adapt to user feedback and context. The model supports multi-shot interactions and handles follow-up questions, representing a significant advancement over static explanation methods by providing contextually appropriate, evolving explanation experiences

The emphasis on process explanations over visualizations challenges the prevailing assumption in HCI that visual representations are universally superior for complex information. This finding suggests that explanation modality preferences may be more context-dependent than previously assumed, raising important questions about the generalizability of visual-first design approaches across different domains and user populations. Consequently, future research should explore hybrid modalities that integrate both visual and process-oriented explanations, or contextually adaptive approaches that dynamically adjust explanation modalities based on user context, task complexity, and individual cognitive capacities. Such adaptive systems could better accommodate diverse user needs, potentially enhancing comprehensibility and user trust across varied interaction scenarios.

4.4.2 Contextual validation

Contextual Validation is a phase where designers refine and verify the interface within the development environment to ensure it aligns with user requirements before full-scale user testing. Four of the 27 reviewed studies provided key insights into this process.

Alvarado et al. (2022) highlighted the importance of iterative testing across diverse scenarios to validate interface usability for group recommendation systems. Through early prototype evaluations, designers identified comprehension gaps and adjusted feature prioritization to better address user needs. Weber and Hußmann (2022) reinforced the role of prototype-based contextual validation in aligning data-driven tools with real-world tasks

Table 5 Synthesis of studies contributing to UI design in XAI system design

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
UI Design	Conceptual Design	Weith and Matt (2023)	Process explanations are more influential than visualizations in shaping user perceptions of fairness in voice agent systems. During the conceptual design phase, it is essential to integrate explanation types that align with user needs, with a particular focus on developing explanations that adapt to individual user profiles.	Pioneered empirical demonstration of process explanation superiority over visualizations for fairness perception	• It is recommended that an iterative co-design process be adopted to continuously refine the interface based on end-user feedback. This approach ensures that explanations align with users' specific contexts and tasks. By involving users in ongoing design adjustments, explanations can be directly shaped by real-world usage and user input, enhancing relevance and usability in dynamic environments (Müller-Birn et al. 2021). • Design EUIs with human-like elements, such as voice feedback and a virtual presence, to make interactions more relatable and intuitive. Adding adaptive visual cues, like shaded confidence intervals to represent uncertainty, can further strengthen trust by making complex information accessible and easy to interpret. This approach is particularly effective for applications where establishing a human-centered connection enhances the AI experience (Weitz et al., 2021; Thomas et al. 2024) • In voice agent design, focus on developing process explanations rather than solely visual cues, as these are more influential on users' perceptions of fairness. Tailoring these explanations to user profiles by considering individual needs and interaction history creates a transparent and equitable experience, fostering trust in AI-driven voice agents. This method, especially with the addition of real-time customization through HDTs, ensures a personalized and fair interaction (Weith and Matt 2023; Song et al., (2023)

Table 5 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
		Thomas et al. (2024)	Incorporating UCD principles when visualizing uncertainty can improve user trust and engagement. The study found that including clear, visual indicators of confidence levels, such as shaded areas, helps users interpret data uncertainty effectively, which is essential for designing trustworthy explanations	Established specific visual uncertainty representation methods that enhance trust	
		Weitz et al. (2021)	Adding human-like elements to virtual agents, such as voice and visual presence, enhances user trust in AI systems. This study suggests that when designing conceptual interfaces, incorporating relatable, human-centered design elements—such as voice feedback and visual representations—can make explanations feel more intuitive and understandable	Quantified trust improvements from incorporating voice feedback and visual presence	
		Müller-Birn et al. (2021)	The case studies were situated within a human-centered design framework, with the objective of creating Explanation User Interfaces (EUI) that reflected the specific needs and contexts of the users in question. The value of iterative co-design and reflective approaches was emphasized, as these facilitate customization based on the tasks and contexts of the users in question.	Established framework for creating EUI reflecting specific user needs	
		Song et al. (2023)	The implementation of Human Digital Twins (HDTs) for customized interface design is a promising avenue for enhancing user experience. By integrating a user's digital twin with the interface, real-time adjustments can be made based on personal and contextual data, leading to more precise explanations and increased user engagement.	Pioneered HDT application to XAI interface personalization	
	Contextual Validation	Alvarado et al. (2022)	The necessity for iterative testing with diverse scenarios to validate the usability of interfaces within group recommendation systems was underscored. The evaluation of early-stage prototypes enabled designers to address deficiencies in system comprehension and optimize the prioritization of features.	Established methodology for early-stage prototype evaluation in collaborative systems	• Scenario-based tests during prototype validation help assess explanation and model efficacy in real-world contexts. Specific scenarios, like "person near traffic light" in semantic segmentation, enable precise evaluation of model accuracy and reveal performance issues, even without labeled datasets (Hoque et al., 2023; Alvarado et al. 2022).

Table 5 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
					• Mixed-reality or simulation platforms enable testing of prototypes under realistic conditions. In applications like automated vehicles, this approach helps fine-tune explanation timing and content to manage cognitive load, improving user deployment and safety (Kim et al. 2023). • Iterative prototype validation allows designers to address usability issues early and prioritize features based on user tasks and expectations. This approach reduces mismatches between system outputs and user needs, enhancing overall interface effectiveness and satisfaction, especially in complex systems like group recommendations (Alvarado et al. 2022; Weber and Hußmann 2022).
		Weber and Hußmann, (2022)	In a survey of data-driven tool development, this research underscored the significance of prototype-based contextual validation to guarantee that data models align with user needs in actual operational contexts. This validation process assesses tool usability and ensures alignment with the tasks and expectations of the intended user.	Formalized approach to validating data-driven tools against operational requirements	
		Hoque et al. (2023)	The study demonstrated that the creation of scenarios with visual concepts is an effective method for contextual validation of semantic segmentation models. The creation of specific test scenarios (e.g., person near traffic light) revealed how model accuracy can be assessed under different visual contexts, thus enabling the identification of specific performance issues without the need for labeled datasets.	Developed novel visual concept scenario method for semantic segmentation validation	

Table 5 (continued)

Stage	Element	Study	Relevant findings	Major contribution	Practical knowledge/insights
		Kim et al. (2023)	A mixed-reality driving platform was developed to test the efficacy of visual explanations in automated vehicles under realistic conditions. The validation method permitted the adjustment of explanation timing and content, thereby optimizing it to reduce passenger cognitive load prior to user deployment.	Demonstrated application of mixed-reality validation for automated vehicle explanations	

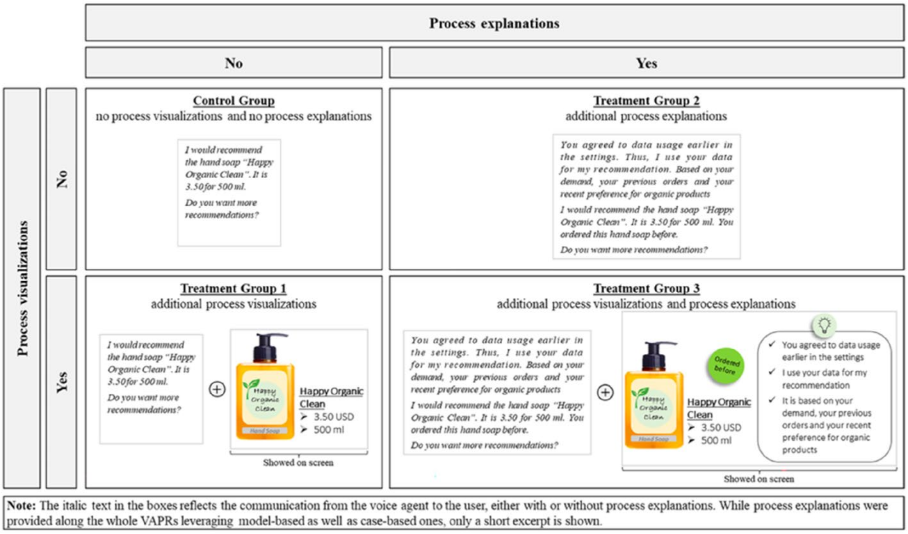

Fig. 8 Experimental Design Comparing Process Explanations and Visualizations in Voice Agent Systems. (Adapted from Weith and Matt 2023). The study design evaluates different explanation modalities' impact on fairness perceptions

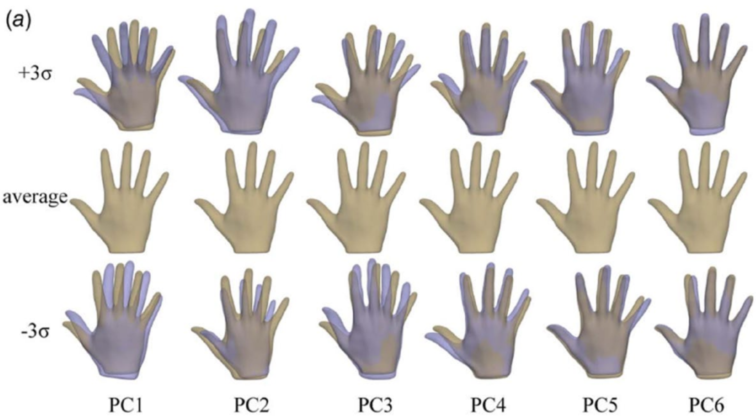

Fig. 9 Principal Component Analysis (PCA) Factors for Human Digital Twin (HDT) Personalization. (Adapted from Song et al., 2023). The visualization demonstrates HDT technology application for creating dynamically personalized explanation interfaces

and expectations, ensuring models function effectively within actual operational contexts. Hoque et al. (2023) demonstrated that specific visual scenarios, such as "person near traffic light" in semantic segmentation models, can expose performance issues without requiring labeled datasets, enabling a more precise assessment of model accuracy across varied conditions. Kim et al. (2023) leveraged a mixed-reality driving platform to test visual explanations in automated vehicles, allowing developers to fine-tune explanation timing and content to reduce cognitive load and enhance user experience before larger-scale deployment.

Several practical insights emerge from these findings for Contextual Validation. First, scenario-based testing in prototype validation helps designers evaluate explanation and model effectiveness under realistic conditions, uncovering crucial performance gaps and enhancing model reliability, even without extensive labeled data (Hoque et al., 2023; Alvarado et al. 2022). Second, using mixed-reality and simulation platforms for testing in lifelike environments is especially beneficial for high-stakes applications, such as automated vehicles, enabling precise adjustments to explanation timing and content, optimizing cognitive load management, user safety, and engagement during deployment (Kim et al. 2023). Finally, an iterative prototype validation process allows designers to address usability issues early, ensuring feature prioritization is closely aligned with user tasks and expectations, minimizing discrepancies between system outputs and user requirements, ultimately improving interface effectiveness and user satisfaction in complex systems like group recommendations (Alvarado et al. 2022; Weber and Hußmann 2022).

4.5 Evaluation and refinement

The final stage of our UCXAISD framework, Evaluation and Refinement, focuses on ensuring system effectiveness through comprehensive assessment and continuous improvement. Through usability testing with end users and systematic refinement based on feedback, this stage evaluates three critical aspects of XAI systems: effectiveness, usability, and trustworthiness. Our analysis of the literature reveals two key components within this stage: Usability Testing and Iterative Improvement. Table 6 synthesizes current research contributions to this stage, providing evidence-based insights into XAI system evaluation and refinement practices.

4.5.1 Usability testing

Usability testing helps identify design elements that improve user trust, comprehension, and interaction. Fan et al. (2020) demonstrated that a visual analytics tool augmented with machine learning could efficiently detect usability issues in think-aloud sessions by automatically highlighting problem areas in recorded interactions, providing deeper insights for UX practitioners. In testing virtual agents, Weitz et al. (2021) found that adding anthropomorphic features, like voice feedback and visual presence, fosters user trust and relatability, making explanations more intuitive. Similarly, Sepich et al. (2023) emphasized usability testing's role in optimizing human-agent collaboration through real-time, interactive explanations, which improve users' understanding and engagement with complex AI-driven explanations. Finally, Thomas et al. (2024) showed that transparent visualizations of uncertainty in active learning systems enhance user trust and decision-making, emphasizing the importance of clear, accessible displays of AI confidence.

Several practical insights emerge from these findings. Integrating machine learning-powered visual analytics into usability testing enhances efficiency by automatically identifying usability issues in recorded user sessions, giving UX practitioners a precise view of user challenges and streamlining the feedback process (Fan et al. 2020; Thomas et al. 2024). Additionally, in testing virtual agents, including human-like features such as voice feedback and contextually adaptive, real-time responses enhances user trust and engagement with AI explanations. This approach promotes natural, intuitive interactions, supporting both user comprehension and effective collaboration in complex, interactive settings (Weitz et al., 2021; Sepich et al. 2023).

The reliance on traditional usability testing methods across all reviewed studies (Fan et al. 2020; Weitz et al., 2021; Sepich et al. 2023) reveals a potential blind spot in XAI evaluation. Unlike conventional interfaces, XAI systems fundamentally alter users' understanding and decision-making processes, yet current evaluation approaches fail to capture these cognitive changes systematically. This limitation suggests that the field requires new evaluation paradigms that can assess explanation impact on user mental models and long-term decision quality.

4.5.2 Iterative improvement

Iterative improvement ensures explanations and interfaces are consistently aligned with evolving user needs and operational contexts. Weber and Hußmann (2022) highlighted the necessity of iterative feedback in refining data-driven tools, emphasizing its importance in bridging the gap between tool functionality and practical usability within complex environments. Similarly, Grafe et al. (2022) found that iterative adaptation in in-vehicle interfaces, informed by real-time driver feedback, helped maintain transparency and control as driving conditions changed, thereby meeting dynamic user expectations. Muntwiler and Eppler (2023) further illustrated the value of iterative feedback in adjusting explanations for users by countering the "illusion of explanatory depth" with visual knowledge calibration, ultimately supporting more accurate decision-making in complex scenarios.

Informed by these findings, iterative improvement in UCXAISD should focus on establishing a feedback loop that continuously refines explanations and data models to remain responsive to user requirements and situational demands (Weber and Hußmann 2022). Adaptive interfaces particularly benefit from real-time feedback, enabling immediate adjustments that address comprehension gaps, such as users' overestimations of their understanding of AI outputs. This iterative process not only enhances transparency and usability but also builds user trust and supports effective decision-making in complex, data-driven environments (Grafe et al., 2022; Muntwiler and Eppler 2023).

5 Discussion and conclusion

The growing demand for transparency and trust in AI-driven decision-making has made UCXAISD increasingly crucial. Our systematic review of 27 studies synthesizes existing knowledge into a comprehensive framework with five key stages: Contextual Inquiry, Explanation Needs Identification, XAI Method Selection, UI Design, and Evaluation and Refinement. For each stage, we provide evidence-based guidelines that serve as a blueprint for developing XAI systems prioritizing user needs and cognitive capabilities.

Our framework successfully extends traditional UCD processes by incorporating specialized considerations for XAI systems. As demonstrated in Fig. 10, we bridge the general UCD stages (Research, Conceptual Design, Prototyping, User Testing, and Iterative Refinement) with XAI-specific elements. This integration addresses critical gaps in traditional UCD approaches by emphasizing explainability and interactivity in AI systems. The framework incorporates essential elements identified by Green and Labrecque (2023), including

Table 6 Synthesis of studies contributing evaluation and refinement in XAI system design

Stage	Element	Study	Relevant Findings	Major Contribution	Practical Knowledge/Insights
Evaluation and Refinement	Usability Testing	Fan et al. (2020)	A visual analytics tool for analyzing think-aloud sessions identified more usability issues than traditional methods, using machine learning to pinpoint problem areas in recorded sessions and enabling UX practitioners to detect issues more efficiently and with greater insight.	Pioneered machine learning integration for automatic detection of problem areas in UX sessions	• The utilization of visual analytics tools augmented with machine learning facilitates the examination of think-aloud sessions and recorded user interactions. This methodology enables UX practitioners to discern usability issues with greater efficacy and precision, as it automatically identifies problematic areas within session recordings, thereby providing insights into the locations and causes of user difficulties (Fan et al. 2020; Thomas et al. 2024). • It is recommended that anthropomorphic features, such as voice feedback and visual presence, be included in usability tests for virtual agents. This is intended to increase user trust and relatability. In scenarios involving interactive explanations, it is advised that real-time, contextually adaptive feedback be given priority. This is to support effective collaboration and enhance user understanding. This approach ensures that users can engage naturally with complex AI systems while maintaining a clear, contextual understanding of explanations (Weitz et al., 2021; Sepich et al. 2023).
		Weitz et al. (2021)	Usability testing of virtual agents with anthropomorphic features, such as voice feedback and visual presence, was found to enhance user trust and relatability of explanations. This testing ensures that human-centered design elements effectively support natural interactions with XAI systems.	Quantified trust improvements from human-like elements in virtual agents	

Table 6 (continued)

Stage	Element	Study	Relevant Findings	Major Contribution	Practical Knowledge/Insights
		Sepich et al. (2023)	The study highlighted usability testing as essential for identifying design elements that facilitate effective human-agent collaboration, especially in real-time, interactive explanations. This testing helps ensure that users can understand and engage with complex explanations, crucial for performance in collaborative environments.	Established methodology for testing interactive explanations in collaborative environments	
		Thomas et al. (2024)	Usability testing in active learning systems revealed that transparent visualizations of uncertainty improve user trust and decision-making. Testing demonstrated the importance of classifier confidence clarity in empowering users to make informed choices without over-relying on AI.	Empirically evaluated demonstrating how specific uncertainty visualization techniques directly impact user trust calibration and decision quality in active learning systems	
	Iterative Improvement	Weber and Hußmann (2022)	Iterative feedback was essential for refining data models in data-driven tools, ensuring they remained aligned with user tasks and operational demands. This approach bridged the gap between tool functionality and practical usability in complex environments.	Formalized iterative refinement approach for data-driven tools in complex environments	• To maintain alignment with user needs and operational contexts, establish an iterative feedback process that continuously refines explanations and data models. Regularly gather and incorporate user feedback to adjust explanations, ensuring transparency and usability in dynamic environments. For adaptive interfaces, prioritize real-time feedback loops to address situational demands and adjust for comprehension gaps, such as the "illusion of explanatory depth." This iterative approach enhances user understanding of AI outputs and improves decision-making quality in complex, data-driven applications (Weber and Hußmann 2022; Grafe et al., 2022; Muntwiler and Eppler 2023).

Table 6 (continued)

Stage	Element	Study	Relevant Findings	Major Contribution	Practical Knowledge/Insights
		Grafe et al. (2022)	The study demonstrated that iterative adaptation of in-vehicle interfaces, based on real-time driver feedback, helps maintain transparency and control in changing conditions. This process allowed the interface to better meet dynamic user expectations and situational needs.	Demonstrated dynamic interface adaptation based on changing conditions	
		Muntwiler and Eppler (2023)	Using visual knowledge calibration techniques, this study applied iterative feedback to adjust explanations and counter the "illusion of explanatory depth." This method improved users' understanding of AI outputs, leading to more accurate decision-making in complex scenarios.	Developed novel approach to improve user comprehension of complex AI outputs	

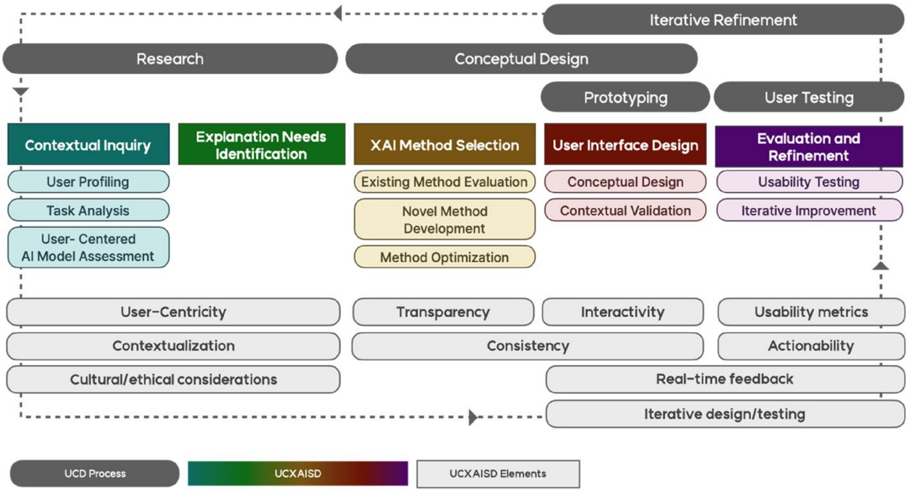

Fig. 10 Proposed UCXAISD Framework Aligned with Traditional UCD Process and XAI Elements. The framework integrates established UCD principles with XAI-specific considerations, mapping UCXAISD stages against traditional UCD phases while incorporating critical elements like transparency, actionability, and user-centricity

user-centricity, contextualization, transparency, and real-time feedback, which are fundamental for fostering user acceptance and trust (Fig. 10).

Beyond these core contributions, our review reveals significant trends and gaps in the field. Notable advances include the development of adaptive explanation methods and multi-modal interfaces that better accommodate diverse user needs and interaction preferences. However, the research landscape remains fragmented, with limited integration of user requirements into critical stages such as XAI method selection and interface design. While technical advancements in explainability methods are abundant, fewer studies address how these methods align with real-world user requirements, particularly in high-stakes domains like healthcare and autonomous systems.

Our systematic review reveals several critical challenges that continue to impede progress in user-centered XAI design. The most prominent challenge is the persistent tension between technical accuracy and human comprehension, where highly performing complex models often resist intuitive explanation, while interpretable models may sacrifice predictive power. Additionally, the field struggles with standardized evaluation methodologies for measuring explanation effectiveness, as current metrics often fail to capture the subjective and context-dependent nature of user understanding and trust.

Despite these challenges, several promising trends are emerging from our analysis that suggest the field's evolution toward more sophisticated and user-responsive systems. There is a clear shift toward personalization and adaptive explanation systems, exemplified by the integration of HDT and dynamic behavior tree models that adjust explanations in real-time based on user feedback and contextual demands. Interactive and conversational XAI approaches are gaining prominence, moving beyond static explanations toward dialogue-based systems that support iterative user inquiry and clarification. However, critical research gaps persist, particularly in developing standardized frameworks for cross-domain evaluation, conducting longitudinal studies to understand how user-XAI relationships evolve over time, and creating practical implementation guidelines that bridge the gap between research insights and real-world deployment.

A particularly significant gap exists in addressing the needs of underrepresented populations. Researchers like Pradeep (2023) and Oswal (2019) emphasize the importance of involving users with disabilities in the design process, underscoring the necessity for a more comprehensive, inclusive approach to XAI system design. While previous research has proposed UI and UX design guidelines for people with developmental disabilities (Hong et al. 2024a, 2024b) and low-literate users (Srivastava et al. 2021), there remains a critical need for focused research on explanation design tailored to these populations within AI systems. Accessibility and inclusivity in XAI system design remain insufficiently addressed, despite their critical role in ensuring broad adoption and usability. Future research should incorporate accessibility frameworks such as WCAG (Web Content Accessibility Guidelines) and explore how adaptive explanations can accommodate individuals with cognitive impairments, vision impairments, and low digital literacy. For example, users with cognitive disabilities may benefit from simpler explanation structures with visual supports, while users with vision impairments require non-visual explanation alternatives that maintain the same level of detail and user agency. Integrating inclusive design principles will enhance the practical applicability of XAI systems across diverse user groups and ensure that AI transparency does not inadvertently create new forms of digital exclusion.

Another underexplored aspect is how cultural differences influence explanation needs. User expectations regarding AI transparency, trust, and interpretability may vary significantly based on cultural backgrounds, impacting how explanations should be structured. For instance, preferences for implicit versus explicit explanations, reliance on authority-based decisions, or different cognitive processing styles can shape how users engage with XAI systems. Cultural dimensions such as power distance, uncertainty avoidance, and collectivism versus individualism may influence whether users prefer detailed technical explanations or higher-level conceptual frameworks. Future work should investigate cross-cultural variations in explainability needs and test explanation frameworks across different cultural groups to ensure global applicability of XAI systems.

It is important to acknowledge the limitations of our study. Despite comprehensive coverage, the rapid evolution of the field means some relevant research may have been missed. Additionally, one key limitation is the reliance on existing literature without empirical validation of the proposed guidelines. While the systematic review provides a rigorous synthesis of user-centered XAI design principles, it does not assess their practical effectiveness through real-world implementation or user studies. To strengthen the applicability of these guidelines, future research should focus on conducting empirical evaluations, such as usability testing, A/B testing, and longitudinal studies, to measure the impact of these guidelines on user interaction, trust, and decision-making in XAI systems. In particular, the use of AI testbeds and benchmarking strategies as outlined above would provide a structured path toward empirical validation, allowing researchers to systematically compare different explainability approaches and refine best practices for responsible AI development (Raza et al. 2025). For example, comparative studies could evaluate how different explanation types (e.g., feature attribution versus counterfactual explanations) affect user trust and decision quality across different task contexts. Additionally, incorporating case studies of deployed XAI interfaces would provide concrete evidence supporting the framework's efficacy and reveal potential refinements necessary for different application domains. Such empirical validation is particularly important for understanding how the proposed framework performs across diverse domains with varying levels of risk and user expertise requirements.

In conclusion, this systematic review provides a valuable foundation for UCXAISD, offering a comprehensive and structured process and guidelines that bridge fragmented research and practical implementation. By following these guidelines, researchers and practitioners can create XAI systems that not only provide technically sound explanations but also ensure these explanations are meaningful and accessible to intended users across various domains. Future research should prioritize empirical validation of proposed process and guidelines and development of adaptive XAI systems that cater to diverse user needs, thereby enhancing the overall accessibility and effectiveness of AI technologies.

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Author contributions S.H. designed the research, conducted literature review and analysis, and wrote the main manuscript text. W.P. participated in literature analysis and provided research direction, overall supervision, and revised the manuscript. All authors reviewed and approved the final manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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